SOLUTIONS: ESTIMATION AND FAULT DETECTION

1. Solution

a) Many ways exist to obtain a transfer function. As a fast way, the transfer function $G_{yr}(s)$ can be obtained as follows (by using well-known block-algebra rules):

$$G_{yr}(s) = \frac{G_1(s)G_2(s)G_3(s)G_4(s) + G_1(s)G_4(s)}{1 + KG_1(s)G_2(s)G_3(s)G_4(s) + KG_1(s)G_4(s) - G_3(s)G_4(s)}.$$

Typical mistakes include not realising that block algebra can be conveniently used to reduce the computational burden.

[3 Marks]

b) Based on the closed-loop block-scheme of the system, it can be said that

$$(R(s) - X_3(s))\frac{1}{s+5} = X_1(s),$$
$$(5X_1(s) + X_3(s))\frac{1}{s} = X_2(s),$$
$$(X_1(s) + X_2(s))\frac{1}{s+3} = X_3(s)$$

yielding

$$\dot{x}_1 = -5x_1 - x_3 + r,$$
$$\dot{x}_2 = 5x_1 + x_3,$$
$$\dot{x}_3 = x_1 + x_2 - 3x_3.$$

Therefore, the system state matrix and output matrix are as follows:

$$A = \begin{bmatrix} -5 & 0 & -1 \\ 5 & 0 & 1 \\ 1 & 1 & -3 \end{bmatrix}, B = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, C = \begin{bmatrix} 0 & 0 & 1 \end{bmatrix}.$$

Typical errors: confusing row and column vectors in forming matrices A, B, C, D and misplacing rows and/or columns

[3 Marks]

c) Based on the obtained A and C, the observability can be checked by checking the rank of the observability matrix as follows:

$$\text{rank}\left(\begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}\right) = \text{rank}\left(\begin{bmatrix} 0 & 0 & 1 \\ 1 & 1 & -3 \\ -3 & -3 & 9 \end{bmatrix}\right) = 2.$$

Answers

Moreover, to determine a basis for the nonobservable vector subspace, we should find v such that

$$\begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix} v = 0_{3\times 1}.$$

Therefore,

$$v = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}.$$

Typical mistakes include errors in evaluating the rank of the observability matrix and also failing to construct the non-observability subspace by choosing the correct vectors forming a basis for it.

[3 Marks]

d) To check the unobservable pole, we should find the pole (an eigenvalue of A) not appearing in the output of the system. First we obtain the eigenvalues of A which are the roots of $\det(sI-A)$ as follows:

$$\det(sI-A) = \det \begin{bmatrix} s+5 & 0 & 1 \\ -5 & s & -1 \\ -1 & -1 & s+3 \end{bmatrix} = s(s+3)(s+5).$$

Therefore, the eigenvalues of A are

$$\text{eig}(A) = \{-5,-3,0\}.$$

Moreover, it can be said that

$$(sI-A)^{-1} = \begin{bmatrix} \frac{s^2+3s-1}{s(s+3)(s+5)} & \frac{-1}{s(s+3)(s+5)} & \frac{-1}{(s+3)(s+5)} \\ \frac{5s+16}{s(s+3)(s+5)} & \frac{s^2+8s+16}{s(s+3)(s+5)} & \frac{1}{(s+3)(s+5)} \\ \frac{1}{s(s+3)} & \frac{1}{s(s+3)} & \frac{1}{s+3} \end{bmatrix}.$$

Now, we check the roots appearing in the output as follows:

$$C(sI-A)^{-1} = \begin{bmatrix} \frac{1}{s(s+3)} & \frac{1}{s(s+3)} & \frac{1}{s+3} \end{bmatrix}.$$

Therefore, we can conclude that $s=-5$ is the unobservable pole of the closed-loop system.

The typical error is not realising that $C(sI-A)^{-1}$ is a row-vector and hence the correct definition of pole has to be used.

[4 Marks]

e) We should determine a basis T for the orthogonal complement of $\ker(Q)$ where

$$Q = \begin{bmatrix} C \\ CA \\ CA^2 \end{bmatrix}.$$

Considering the nonobservable vector subspace of Q, the matrix T can be chosen as

$$T = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}.$$

Answers

Now, we consider a similarity transformation matrix T and a new variable z such that $x = Tz$. Therefore, by defining $z = \begin{bmatrix} z_1 & z_2 & z_3 \end{bmatrix}^\top$,

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = T^{-1}ATz + T^{-1}Br,$$
$$y = CTz.$$

It can be said that

$$T^{-1} = \begin{bmatrix} 0.5 & -0.5 & 0 \\ 0 & 0 & 1 \\ 0.5 & 0.5 & 0 \end{bmatrix}.$$

Thus,

$$\begin{bmatrix} \dot{z}_1 \\ \dot{z}_2 \\ \dot{z}_3 \end{bmatrix} = \begin{bmatrix} \mathbf{-5} & -1 & -5 \\ 0 & -3 & 2 \\ 0 & 0 & 0 \end{bmatrix} z + \begin{bmatrix} 0.5 \\ 0 \\ 0.5 \end{bmatrix} r,$$
$$y = \begin{bmatrix} 0 & 1 & 0 \end{bmatrix}$$

in which the subsystem described by $\begin{bmatrix} \dot{z}_2 & \dot{z}_3 \end{bmatrix}^\top$ is observable.

The typical error is in the calculation of the inverse of T

[4 Marks]

f) Output feedback cannot change the observability of the system for any K.

The frequent oversight is not recalling that algebraic output feedback does not change the observability properties

[3 Marks]

Answers

2. Solution

a) Since $R = 1$ and $L = 1$, by considering $x_1 = V_c$ and $x_2 = I$, it can be said that

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= -x_2 - x_1 + u, \\ y &= x_1. \end{aligned}$$

[3 Marks]

b) By defining

$$A = \begin{bmatrix} 0 & 1 \\ -1 & -1 \end{bmatrix}, B = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 \end{bmatrix},$$

the full-order state observer can be designed as follows:

$$\dot{\hat{x}} = A\hat{x} + Bu + L(y - C\hat{x}).$$

The dynamics of e can be stated as follows:

$$\dot{e} = Ae - LCe$$

implying

$$F = A - LC.$$

By defining $L = \begin{bmatrix} l_1 \\ l_2 \end{bmatrix}$, F is as follows:

$$F = \begin{bmatrix} -l_1 & 1 \\ -1 - l_2 & -1 \end{bmatrix}.$$

Therefore, it is easy to say that if $l_1 = 1$ and $l_2 = -1$, the eigenvalues of F are as -1 and -1.

Typical mistake is setting the incorrect constraint on the desired eigenvalues

[4 Marks]

c) From the Laplace transformation of $\dot{e} = Fe$, it can be said that

$$(sI - F)E(s) = e(0).$$

Thus,

$$e(t) = \mathscr{L}^{-1}\{(sI - F)^{-1}\}e(0)$$

which based on F, can be obtained as

$$e(t) = \mathscr{L}^{-1}\left\{ \begin{bmatrix} s+1 & -1 \\ 0 & s+1 \end{bmatrix}^{-1} \right\} e(0).$$

Therefore,

$$e(t) = \mathscr{L}^{-1}\left\{ \begin{bmatrix} \frac{1}{s+1} & \frac{1}{(s+1)^2} \\ 0 & \frac{1}{s+1} \end{bmatrix} \right\} e(0)$$

and

$$e(t) = \begin{bmatrix} e^{-t} & te^{-t} \\ 0 & e^{-t} \end{bmatrix} e(0).$$

Answers

As a result

$$\varepsilon(t) = Ce(t) = \begin{bmatrix} e^{-t} & te^{-t} \end{bmatrix} e(0).$$

Typical errors are made in the calculation of $\mathscr{L}^{-1}\{(sI-F)^{-1}\}$.

[5 Marks]

d) By considering answer to Question 2-b), in the presence of faults in $u(t)$, we have

$$\dot{e} = Fe + Bf.$$

It can be said that

$$(sI-F)E(s) = e(T_0) + B\frac{f}{s}.$$

Thus,

$$e(t) = \mathscr{L}^{-1}\{(sI-F)^{-1}\}e(T_0) + \mathscr{L}^{-1}\left\{(sI-F)^{-1}B\frac{f}{s}\right\}$$

which based on F, can be obtained as

$$e(t) = \mathscr{L}^{-1}\left\{\begin{bmatrix} s+1 & -1 \\ 0 & s+1 \end{bmatrix}^{-1}\right\}e(T_0) + \mathscr{L}^{-1}\left\{\begin{bmatrix} s+1 & -1 \\ 0 & s+1 \end{bmatrix}^{-1} B\frac{f}{s}\right\}.$$

Therefore,

$$e(t) = \mathscr{L}^{-1}\left\{\begin{bmatrix} \frac{1}{s+1} & \frac{1}{(s+1)^2} \\ 0 & \frac{1}{s+1} \end{bmatrix}\right\}e(T_0) + \mathscr{L}^{-1}\left\{\begin{bmatrix} \frac{1}{s(s+1)} & \frac{1}{s(s+1)^2} \\ 0 & \frac{1}{s(s+1)} \end{bmatrix}\right\}Bf$$

and

$$e(t) = \begin{bmatrix} e^{-t} & te^{-t} \\ 0 & e^{-t} \end{bmatrix} e(0) + \begin{bmatrix} 1-e^{-t} & 1-e^{-t}-te^{-t} \\ 0 & 1-e^{-t} \end{bmatrix} Bf.$$

As a result,

$$\varepsilon(t) = Ce(t) = \begin{bmatrix} e^{-t} & te^{-t} \end{bmatrix} e(0) + (1-e^{-t}-te^{-t})f.$$

If we set $\hat{x}(0) = 0$, since $|x(0)| < 5$, we have $|e(0)| < 5$. In this condition, since $|e(0)| < 5$ and $|e^{-t}[1 \;\; t]| = e^{-t}\sqrt{1+t^2}$, by letting

$$\tilde{\varepsilon}(t) = 5e^{-t}\sqrt{1+t^2},$$

a fault is detected if $|\varepsilon(t)| > \tilde{\varepsilon}(t)$.

Frequent mistake is determining the wrong time-evolution of the residual $\varepsilon(t)$ and of its upper bound

[4 Marks]

e) We should define the fault signal as a new state $x_3 = f$ such that since $u_i = V_i + f$, the system state space equation can be restated as follows:

$$\begin{aligned} \dot{x}_1 &= x_2, \\ \dot{x}_2 &= -x_2 - x_1 + V_i + x_3, \\ \dot{x}_3 &= 0, \\ y &= x_1. \end{aligned}$$

Therefore, the modified state matrix, input matrix, and output matrix can be defined as

$$\bar{A} = \begin{bmatrix} 0 & 1 & 0 \\ -1 & -1 & 1 \\ 0 & 0 & 0 \end{bmatrix}, \bar{B} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \bar{C} = \begin{bmatrix} 1 & 0 & 0 \end{bmatrix}. \tag{2.1}$$

Answers

Now, the observability can be checked by checking the rank of the observability matrix as follows:

$$\operatorname{rank}\left(\begin{bmatrix}\bar{C}\\ \bar{C}\bar{A}\\ \bar{C}\bar{A}^2\end{bmatrix}\right) = \operatorname{rank}\left(\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ -1 & -1 & 1\end{bmatrix}\right) = 3.$$

Therefore, the fault is observable. Now, it is sufficient to design an observer for the system (2.1) to estimate the state vector $\hat{x} = \begin{bmatrix}\hat{x}_1 & \hat{x}_2 & \hat{x}_3\end{bmatrix}^\top$.

Most common difficulty is in realising how to embed the model of a constant abrupt fault and in adding its state equation to the original ones to establish a comprehensive estimation procedure. The other typical mistake is to try to design the estimator without reading the hint that advise not to attempt to do this.

[4 Marks]

Answers

3. Solution

a) For $a \neq 0$, since $H \neq 0$, according to

$$P(t+1) = F\Big[P(t) - P(t)H^\top \big(V_2 + HP(t)H^\top\big)^{-1} HP(t)\Big]F^\top + V_1,$$

as $t \to \infty$, $P(t)$ converges to a constant quantity.

Typical mistakes include the incorrect use of the asymptotic convergence theorems

[4 Marks]

b) The general algebraic Riccati equation is

$$P(t+1) = F\Big[P(t) - P(t)H^\top \big(V_2 + HP(t)H^\top\big)^{-1} HP(t)\Big]F^\top + V_1, t = 1, 2, \ldots.$$

- For $F = \frac{1}{2}$, $H = \frac{1}{2}$, $V_1 = 1$, and $V_2 = 1$, we can say that the solution $\bar{P}_1$ for $P(t)$ satisfies

$$\bar{P}_1 = \frac{1}{4}\Big[\bar{P}_1 - \frac{1}{4}\big(1 + \frac{1}{4}\bar{P}_1\big)^{-1}\bar{P}_1^2\Big] + 1$$

which can be simplified as follows:

$$16\bar{P}_1 = 4\bar{P}_1 - \big(1 + \frac{1}{4}\bar{P}_1\big)^{-1}\bar{P}_1^2 + 16$$

which can be simplified as

$$\bar{P}_1^2 + 2\bar{P}_1 - 4 = 0.$$

Therefore, the solutions of $\bar{P}_1$ are

$$\bar{P}_{11} = -3.236, \bar{P}_{12} = 1.236$$

which only the positive solution is acceptable.

- For $F = -\frac{1}{3}$, $H = \frac{1}{2}$, $V_1 = 1$, and $V_2 = 1$, we can say that the solution $\bar{P}_2$ for $P(t)$ satisfies

$$\bar{P}_2 = \frac{1}{9}\Big[\bar{P}_2 - \frac{1}{4}\big(1 + \frac{1}{4}\bar{P}_2\big)^{-1}\bar{P}_2^2\Big] + 1$$

which can be simplified as follows:

$$36\bar{P}_2 = 4\bar{P}_2 - \big(1 + \frac{1}{4}\bar{P}_2\big)^{-1}\bar{P}_2^2 + 36$$

which can be simplified as

$$9\bar{P}_2^2 + 23\bar{P}_2 - 36 = 0.$$

Therefore, the solutions of $\bar{P}_2$ are

$$\bar{P}_{21} = -3.651, \bar{P}_{22} = 1.096$$

which only the positive solution is acceptable.

Typical mistake is the choice of a negative-definite solution of the ARE.

[4 Marks]

Answers

c) According to the obtained $\bar{P}$

$$\bar{K} = F\bar{P}H^\top (V_2 + H\bar{P}H^\top)^{-1}$$

which for $\bar{P}_1 = 1.236$, $F = \frac{1}{2}$, $H = \frac{1}{2}$, and $V_2 = 1$,

$$\bar{K}_1 = 0.2361$$

and for $\bar{P}_2 = 1.096$, $F = -\frac{1}{3}$, $H = \frac{1}{2}$, and $V_2 = 1$,

$$\bar{K}_2 = -0.1434.$$

Therefore, the difference equation yielding the one-step ahead optimal steady-state Kalman prediction is as follows:

$$\hat{x}(t+1|t) = \frac{1}{2}\hat{x}(t|t-1) + 0.2361\big(y(t) - \frac{1}{2}\hat{x}(t|t-1)\big).$$

and

$$\hat{x}(t+1|t) = -\frac{1}{3}\hat{x}(t|t-1) - 0.1434\big(y(t) - \frac{1}{2}\hat{x}(t|t-1)\big).$$

A frequent mistake consists in adding the process and sensor noise in the predictor.

[3 Marks]

d) According the system model and the Kalman filter difference equation, the dynamical system is sketched in Fig. 3.1

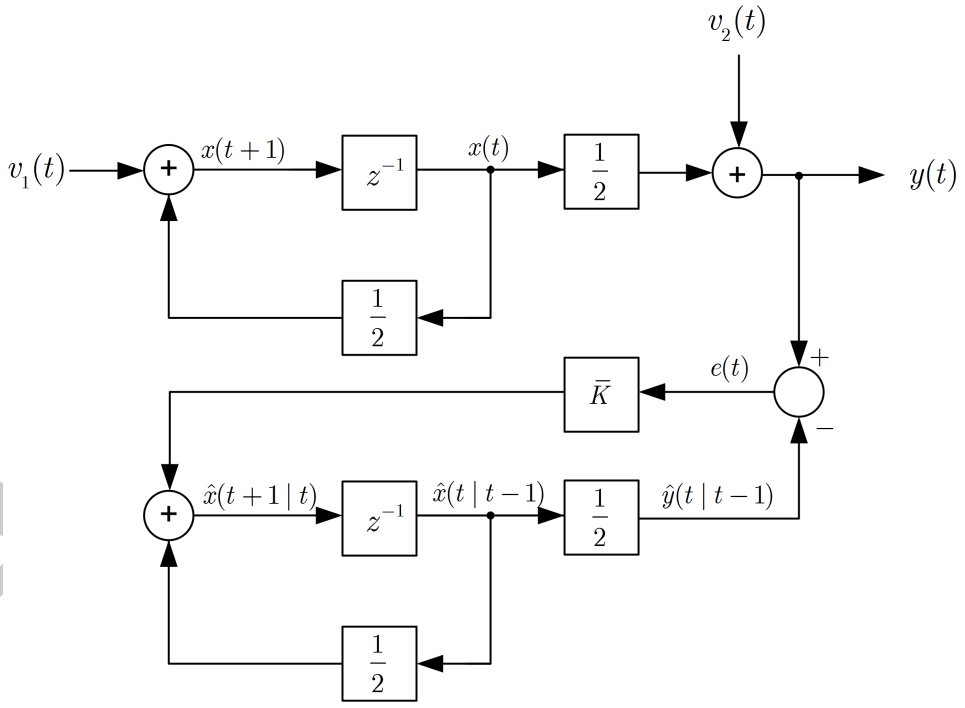


Figure 3.1 The system configuration in Question 3.

This question is very easy to answer.

[3 Marks]

e) The transfer functions from the noises $v_2(t)$ to the estimation error $e(t) = y(t) - \frac{1}{2}\hat{x}(t|t-1)$ are as follows:

$$G^{(2)}_{v_2e} = \frac{Y(z)}{V_2(z)}.\frac{E(z)}{Y(z)} = 1.\frac{1}{1+\frac{1}{2}K\frac{z^{-1}}{1-az^{-1}}} = 1.\frac{1}{1+\frac{K}{2z-2a}}.$$

or

$$G^{(2)}_{v_2e} = \frac{Y(z)}{V_2(z)}.\frac{E(z)}{Y(z)} = \frac{2z-2a}{2z-2a+K}.$$

Hence, denoting by $\Gamma_2(\omega)$ the spectrum of $v_2(t)$ and by $\Gamma_e(\omega)$ the spectrum of $e(t)$, one gets:

$$\Gamma_e(\omega) = \|G^{(2)}_{v_2e}(e^{j\omega})\|^2\Gamma_2(\omega)$$

from which a spectral analysis can be carried out to ascertain the spectral characteristics of $e(t)$ as a function of the value of a.

A common mistake is avoiding the use of block-algebra to calculate the transfer functions.

[6 Marks]

Answers

4. Solution

a) Based on the system state-space equation, we have

$$A=\begin{bmatrix}-1 & 1 & 0\\ 0 & 0 & 1\\ -2 & -1 & 0\end{bmatrix}, C=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 1\end{bmatrix}, B=\begin{bmatrix}0\\ 0\\ 1\end{bmatrix}.$$

It can be said that

$$\det(sI-A)=s^3+s^2+s+3.$$

Therefore,

$$(sI-A)^{-1}=\frac{1}{s^3+s^2+s+3}\begin{bmatrix}s^2+1 & s & 1\\ -2 & s(s+1) & s+1\\ -2s & -(s+3) & s(s+1)\end{bmatrix}$$

and

$$C(sI-A)^{-1}=\frac{1}{s^3+s^2+s+3}\begin{bmatrix}s^2+1 & s & 1\\ -2s-2 & s^2-3 & s^2+2s+1\end{bmatrix}.$$

Moreover, the roots of s^3+s^2+s+3 are $\{-1.5747, 0.2874\pm 1.35i\}$. Therefore, all the poles appear in the output, and the system is fully observable.

Moreover,

$$G_{yu}(s)=C(sI-A)^{-1}B=\frac{1}{s^3+s^2+s+3}\begin{bmatrix}1\\ s^2+2s+1\end{bmatrix}.$$

A very common mistake is not answering the question because students do not realise/remember that observability can be checked using the transfer function.

[5 Marks]

b) Since the rank of C is 2, the order of the reduced order state observer is $n-p=3-2=1$.

Very easy question.

[3 Marks]

c) It is not required to estimate x_1 as it is measurable. If the reduced-order state observer estimate x_2, x_3 can be estimated via the following equation:

$$\hat{x}_3=y_2-\hat{x}_2.$$

By defining $x=\begin{bmatrix}x_1 & x_2 & x_3\end{bmatrix}^\top$, let us consider the following state transformation:

$$z=\begin{bmatrix}z_1\\ z_2\end{bmatrix}=Tx=\begin{bmatrix}I_2 & \begin{bmatrix}0\\ 1\end{bmatrix}\\ 0_{1\times 2} & 1\end{bmatrix}x.$$

Therefore, we have

$$\dot{z}=\tilde{A}z+\tilde{B}u,$$
$$y=\tilde{C}z$$

Answers

where

$$\tilde{A} = TAT^{-1} = \begin{bmatrix} -1 & 1 & -1 \\ -2 & -1 & 2 \\ -2 & -1 & 1 \end{bmatrix}, \tilde{B} = TB = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}, \tilde{C} = CT^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}.$$

Therefore, the reduced-order state observer has the following form:

$$\begin{aligned} \dot{\xi} &= F\xi + Hy + Gu, \\ \hat{z}_2 &= \xi + Ly \end{aligned}$$

where $F = -1$ and L, G, and H can be obtained as follows:

$$\begin{aligned} L\tilde{A}_{12} - \tilde{A}_{22} &= -F, \\ L\tilde{A}_{11} + H - \tilde{A}_{21} &= FL, \\ G + L\tilde{B}_1 - \tilde{B}_2 &= 0. \end{aligned} \tag{4.1}$$

According to the definition of z and $\tilde{A}$,

$$\tilde{A}_{11} = \begin{bmatrix} -1 & 1 \\ -2 & -1 \end{bmatrix}, \tilde{A}_{12} = \begin{bmatrix} -1 \\ 2 \end{bmatrix}, \tilde{A}_{21} = \begin{bmatrix} -2 & -1 \end{bmatrix}, \tilde{A}_{22} = \begin{bmatrix} 1 \end{bmatrix}, \tilde{B}_1 = \begin{bmatrix} 0 \\ 1 \end{bmatrix}, \tilde{B}_2 = 1.$$

Therefore, from (4.1), it follows that

$$\begin{aligned} \begin{bmatrix} l_1 & l_2 \end{bmatrix} \begin{bmatrix} -1 \\ 2 \end{bmatrix} - 1 &= 1, \\ \begin{bmatrix} l_1 & l_2 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ -2 & -1 \end{bmatrix} + \begin{bmatrix} h_1 & h_2 \end{bmatrix} - \begin{bmatrix} -2 & -1 \end{bmatrix} &= -\begin{bmatrix} l_1 & l_2 \end{bmatrix}, \\ G + \begin{bmatrix} l_1 & l_2 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} - 1 &= 0 \end{aligned}$$

which has multiple solution. For instance, to satisfy the first equation, if we set $l_1 = -2$ and $l_2 = 0$, to satisfy the other equations, we should have $h_1 = -2$, $h_2 = 1$, and $G = 1$.

Typical and very common mistake is not understanding that the non-uniqueness of the solution is a not an obstacle in concluding the design. As done in the lectures, it is just a matter of choosing one arbitrary option and proceed.

[8 Marks]

d) Based on the system model and the observer, the block-scheme of the system and of the observer is depicted in Fig. 4.1.

Very easy question.

[4 Marks]

Answers

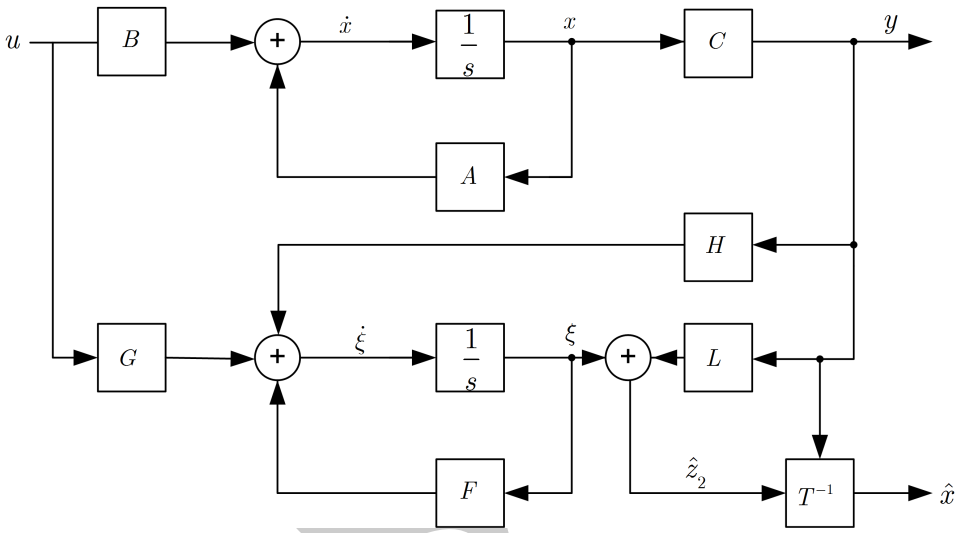


Figure 4.1 The system/observer block-scheme in Question 4-d.

Answers